

CLINICAL INVESTIGATION

Brain

USE OF 3.0-T MRI FOR STEREOTACTIC RADIOSURGERY PLANNING FOR TREATMENT OF BRAIN METASTASES: A SINGLE-INSTITUTION RETROSPECTIVE REVIEW

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Purpose: To investigate the efficacy of 3.0-T magnetic resonance imaging (MRI) for detecting brain metastases for stereotactic radiosurgery (SRS) planning.

Methods and Materials: All adult patients scheduled for SRS treatment for brain metastases at our institution between October 2005 and January 2008 were eligible for analysis. All patients underwent radiosurgery treatment planning 3.0-T MRI on the day of scheduled radiosurgery and a diagnostic 1.5-T MRI in the days or weeks prior to radiosurgery for comparison. Both scans were interpreted by neuroradiologists who reported their findings in the radiology reports. We performed a retrospective review of the radiology reports to determine the number of brain metastases identified using each MRI system.

Results: Of 254 patients scheduled for treatment from October 2005 to January 2008, 138 patients had radiology reports that explicitly described the number of metastases identified on both scans. With a median interval of 17 days (range, 1–82) between scans, the number of metastases detected using 1.5-T MRI system ranged from 1 to 5 and from 1 to 8 using the 3.0 T-MRI system. Twenty-two percent of patients were found to have a greater number of metastases with the 3.0 T-MRI system. The difference in number of metastases detected between the two scans for the entire cohort ranged from 0 to 6. Neither histology ($p = 0.52$ by chi-sq test) nor time between scans ($p = 0.62$ by linear regression) were significantly associated with the difference in number of metastases between scans.

Conclusions: The 3.0-T MRI system appears to be superior to a 1.5-T MRI system for detecting brain metastases, which may have significant implications in determining the appropriate treatment modality. Our findings suggest the need for a prospectively designed study to further evaluate the use of a 3.0 T-MRI system for stereotactic radiosurgery planning in the treatment of brain metastases. © 2010 Elsevier Inc.

Radiosurgery, Brain metastases, MRI.

INTRODUCTION

Brain metastases are diagnosed in approximately 20% of patients with cancer and, if left untreated, portend a median survival of less than 3 months. Early detection and treatment of brain metastases has important implications in both prognosis and quality of life. The appropriate treatment of brain metastases is, in large part, guided by the number of metastases identified. For example, medically operable patients with a solitary metastasis have been traditionally treated with resection followed by whole-brain irradiation. More recent reports, however, demonstrate comparable outcomes in local control when this subset of patients is treated with stereotactic

radiosurgery combined with whole-brain irradiation (1–5). For patients with one to four brain metastases, the treatment options include whole-brain radiotherapy, whole-brain radiotherapy plus radiosurgery, or radiosurgery alone (5–6). In this subset of patients, several randomized trials comparing whole-brain irradiation plus radiosurgery versus radiosurgery alone have found no differences in survival between the two treatment arms, confirming radiosurgery alone as a viable alternative (7, 9–10). For patients with more than four brain metastases, the current treatment of choice at most centers is whole-brain irradiation. At our institution, patients with one to four brain metastases are offered radiosurgery alone as a treatment option, whereas those with

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